

Text Data in Business and Economics

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12. Ethical Considerations

Now, we move from “What is?” to “What ought to be?”

- ▶ Most of the content discussed so far concerns positive questions
 - ▶ How does a given model work
 - ▶ What are we measuring, etc.
- ▶ Today, we will discuss the ethics of text-, image- and audio-data related questions
 - ▶ How should models be built from a social viewpoint
 - ▶ How should we (not) use them, etc.
 - ▶ How should textual, visual or audio data (not) represent our society
- ▶ Ethics discussions are necessarily influenced by moral considerations
 - ▶ What is good or bad? Just or unjust

Scientists do not operate outside of society

- ▶ Positive science is interested in discovery and facts (the “truth” about the world)
- ▶ However, positive science interacts with normative considerations in many ways, for example:
 - ▶ Which questions do we deem important to research
 - ▶ What impact will our discovery have on society
 - ▶ What cost are we willing to carry for discovery (not just monetary)
 - ▶ What responsibilities do we (not) have as scientists
 - ▶ Further examples
- ▶ Values and other context-specific characteristics interact with the entire social science pipeline:
 - ▶ Research question → (data collection) → theoretical or empirical analysis → interpretation

Let us discuss a historic case study: Census data

- ▶ A government wants to establish a comprehensive population registration system for administrative and statistical purposes
- ▶ The system should simplify administration and offer perspectives for social research
- ▶ Questions:
 - ▶ Do you have any ethical concerns
 - ▶ Would you be more/less concerned if it is a company/researchers/another actor constructing the comprehensive database? Why?

History has seen horrible examples of unanticipated secondary use

- ▶ During the Second World War: Dutch census data used to facilitate the genocide against Jews and others:
 - ▶ Data collection during peaceful times
 - ▶ Trusted by citizens
 - ▶ Later misuse by the Nazis
 - ▶ Extremely high death rate of Dutch Jews: 73%
- ▶ More: Seltzer, W., & Anderson, M. (2001). The Dark Side of Numbers: The Role of Population Data Systems in Human Rights Abuses. Social Research, 481-513

Let us discuss further case studies

- ▶ Surveillance technology
 - ▶ Should governments use face recognition technology? If so, for which applications?
 - ▶ How does this question relate to economics research?
- ▶ Representation of different groups (e.g., by ethnicity, gender, age)
 - ▶ From a normative viewpoint, what is the “right” amount and kind of representation of different societal groups in the media
 - ▶ How does this question relate to economics research?
- ▶ Accents and labor market outcomes
 - ▶ (Under what circumstances) should employers be allowed to consider a worker's accent in their hiring decision
 - ▶ How does this question relate to economics research?

The Environmental Impact of LLMs

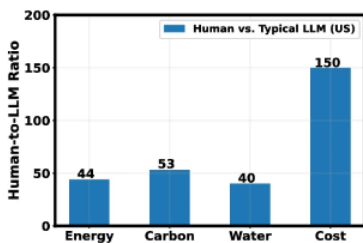
- ▶ LLMs require a significant amount of energy at all stages
 - ▶ ChatGPT is estimated to consume the energy of 33,000 households
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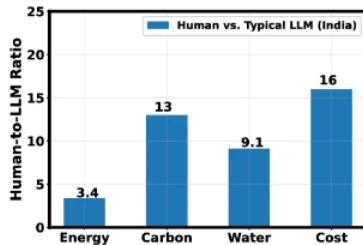
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(a) Typical LLM (U.S.)



(b) Typical LLM (India)

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- ▶ Recent studies have highlighted how LLMs are a more efficient alternative to human labor. However:
 - ▶ LLMs size is growing $\Rightarrow$ energy demand will increase
 - ▶ In the short term, LLMs are unlikely to substitute humans fully

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- ▶ **Users:** Responsible use of GenAI
 - ▶ Revert to Google query for web searches (more reliable and consumes 10 times less than ChatGPT per query)
- ▶ **Companies:** invest in research for *greener* AI systems